

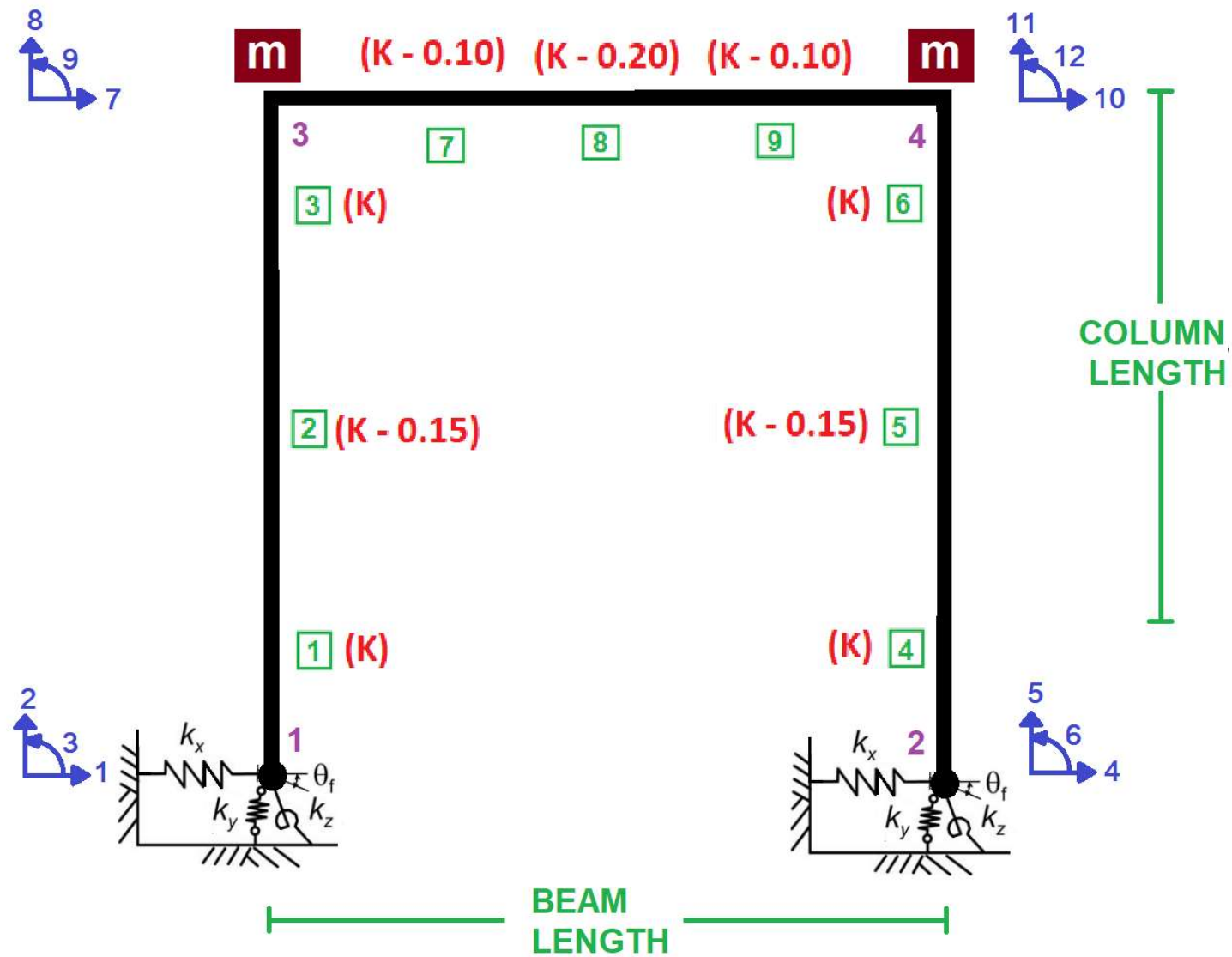
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

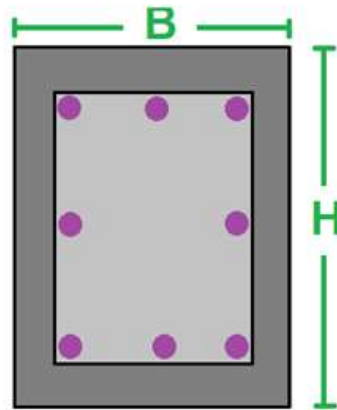
SOIL-FOUNDATION- STRUCTURE INTERACTION USING OPENSEES

SOIL-FOUNDATION INTERACTIONS SIMULATED WITH SIMPLE SPRINGS.

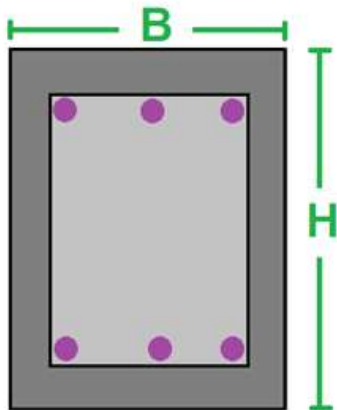
SOIL SPRINGS VALUES ARE NOT EXACT.

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

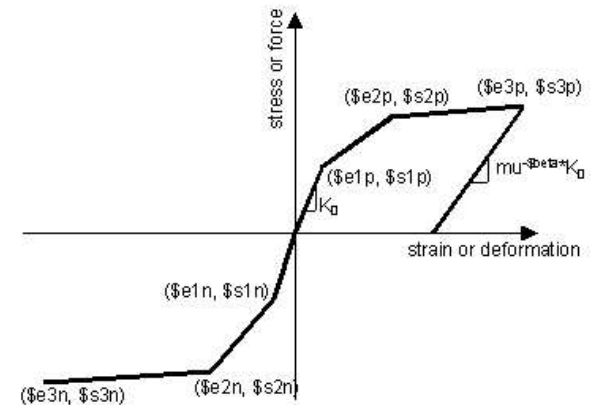
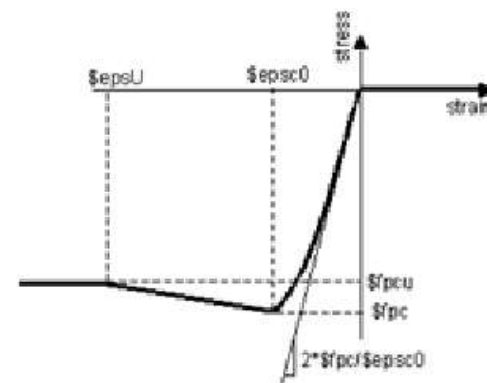
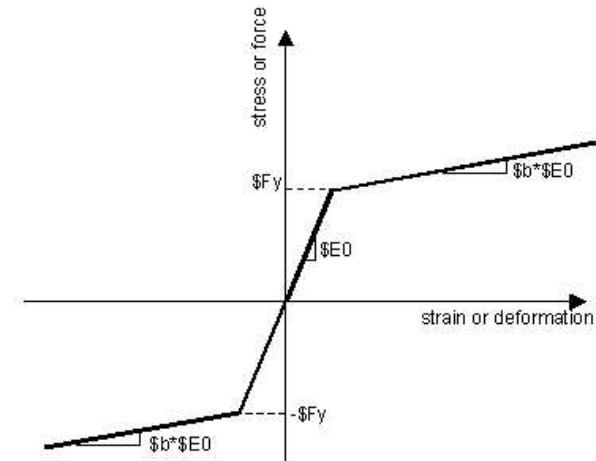
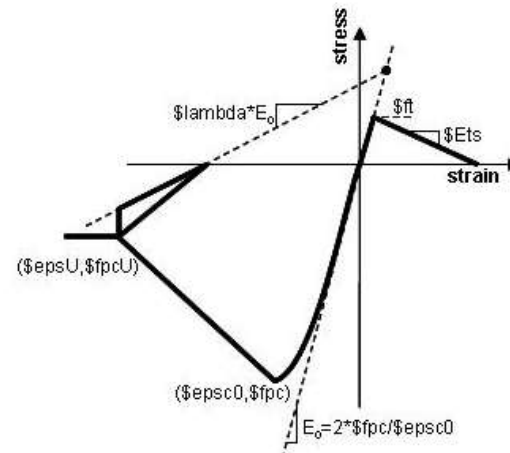




COLUMN SECTION

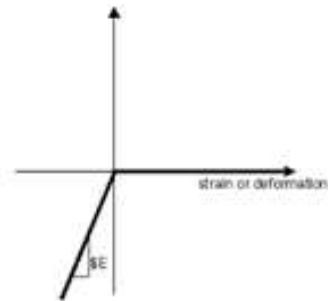


BEAM SECTION

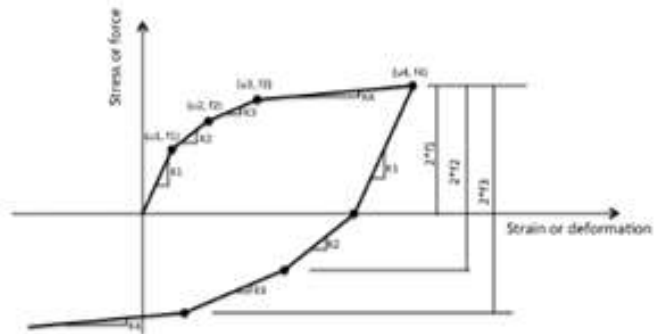




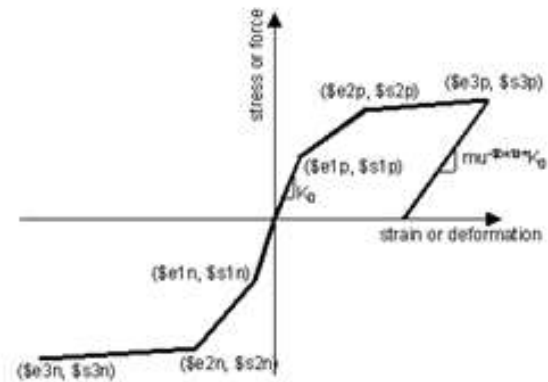
Elastic Uniaxial Material



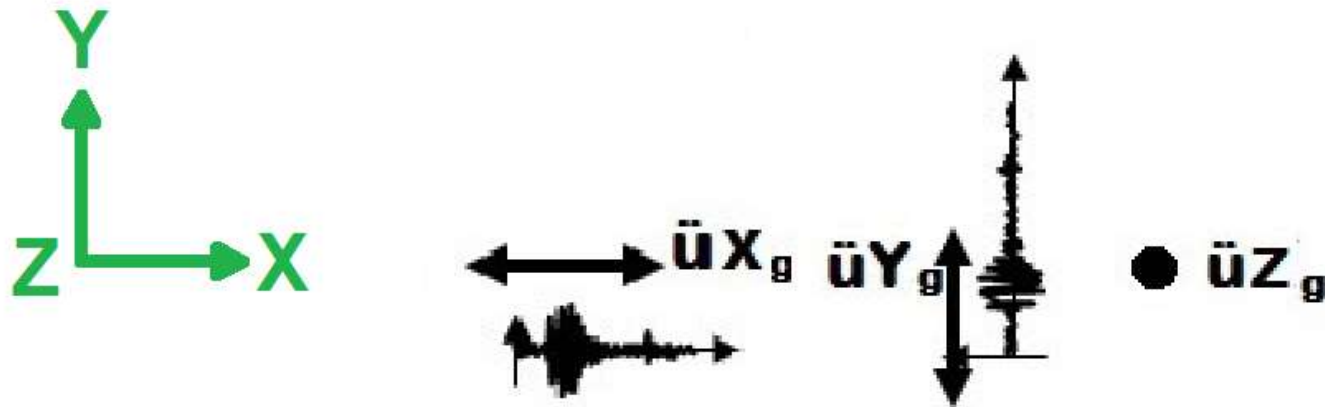
Elastic-No Tension Material



MultiLinear Material



Hysteretic Material



$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displaement from Dynamic Analysis

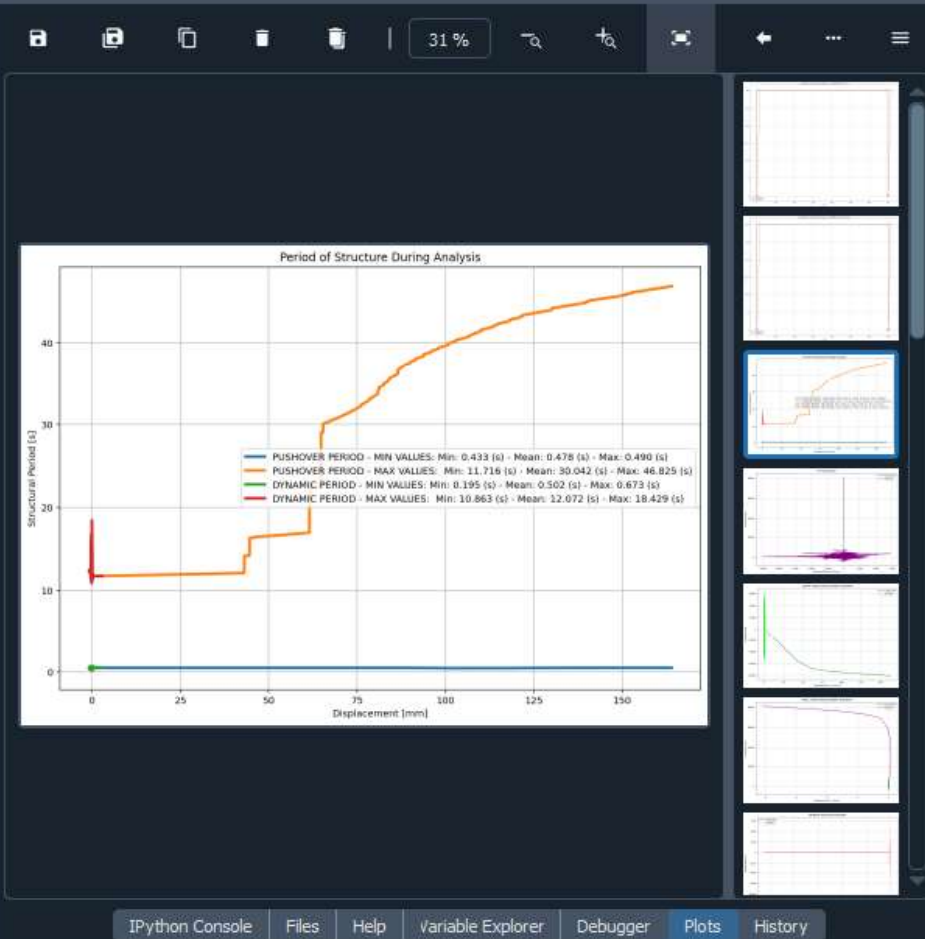
Δ_y = Lateral Yield Displaement from Pushover Analysis

Δ_u = Lateral Ultimate Displaement from Pushover Analysis

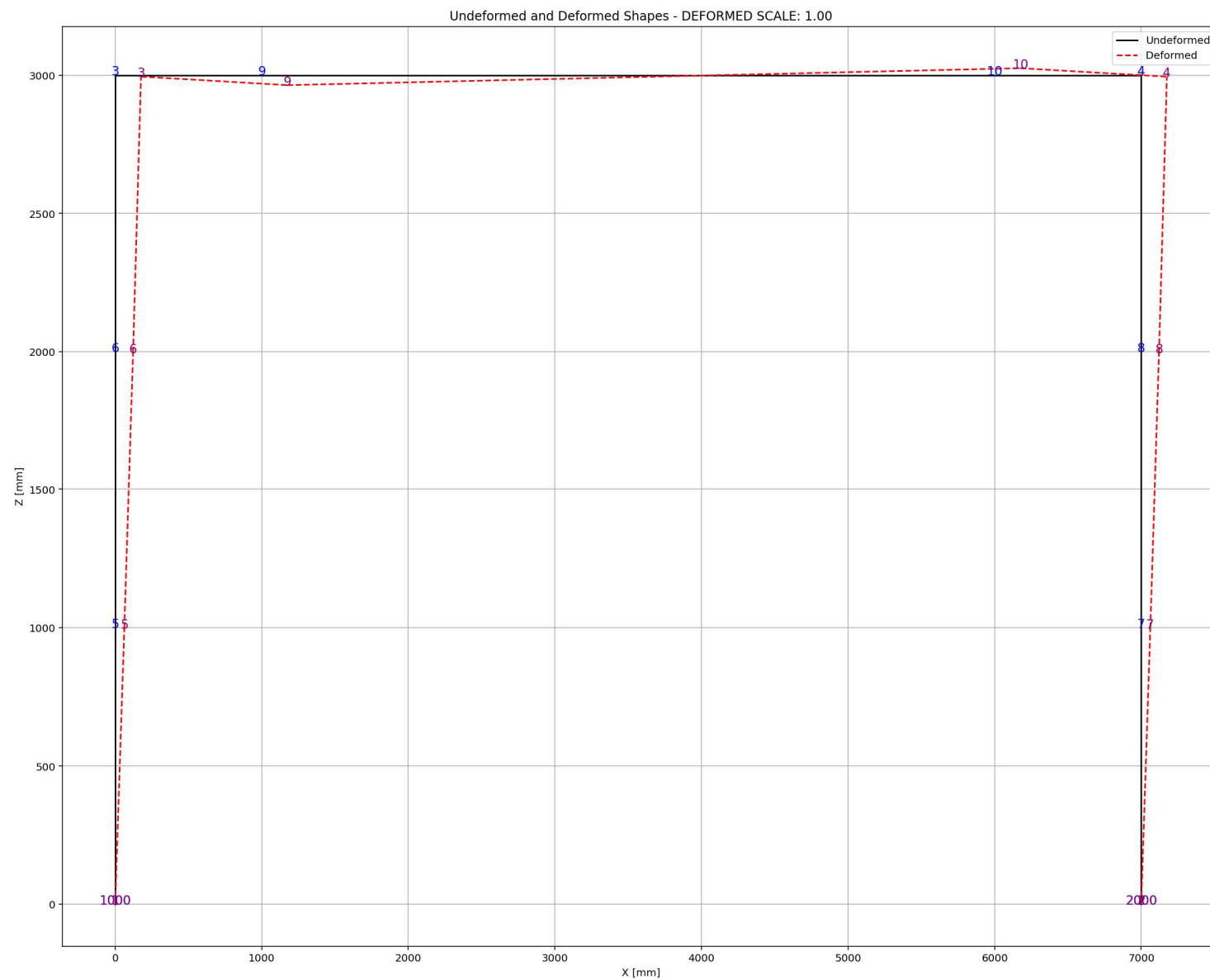
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1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL
3 # SOIL-FOUNDATION-STRUCTURE INTERACTION USING OPENSEES
4 #-----
5 # EXAMPLE 01:
6 # SOIL-FOUNDATION INTERACTIONS SIMULATED WITH SIMPLE SPRINGS.
7 # SOIL SPRINGS VALUES ARE NOT EXACT.
8 #-----
9 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
10 # EMAIL: salar.d.ghashghaei@gmail.com
11 #####
12 """
13 [1] Nonlinear Frame Modeling: 2D RC frame with distributed plasticity (fiber sections) using
14 [2] Material Laws:
15     - *Concrete*: `Concrete01` with confined (core) and unconfined (cover) properties.
16     - *Steel*: `Hysteretic` model with pinching, hardening, and cyclic degradation.
17 [3] Seismic Loads:
18     - Pushover: Displacement-controlled lateral loading to failure.
19     - Dynamic: Uniform excitation with user-defined ground motions (X/Y components).
20 [4] Damping: Rayleigh damping (a0, a1) calibrated via eigenvalue analysis (modes 1-2).
21 [5] Performance Metrics:
22     - Ductility Ratio ( $\mu$ ): Derived from bilinearized pushover curves.
23     - Overstrength ( $O_0$ ): Yield vs. ultimate capacity.
24     - Damage Index (DI): Normalized displacement demand/capacity.
25 [6] Advanced Solver: HHT-a integrator (unconditionally stable) with Newton-Raphson iteration
26 [7] Outputs:
27     - Hysteretic responses (P-M, V- $\Delta$ , M- $\theta$ ).
28     - Time-history plots (displacement, base shear).
29     - Stiffness degradation tracking.
30 [8] Validation: Logarithmic decrement method for damping ratio verification.
31 [9] Ductility Damage Index (DDI) Implementation:
32     DDI quantifies structural damage via normalized displacement demand.
33     # After bilinear fit ( $X[1] = \Delta_y$ ,  $X[2] = \Delta_u$ ):
34      $DD = \max(\text{abs}(\text{DISP } X_d))$  # Max dynamic displacement demand

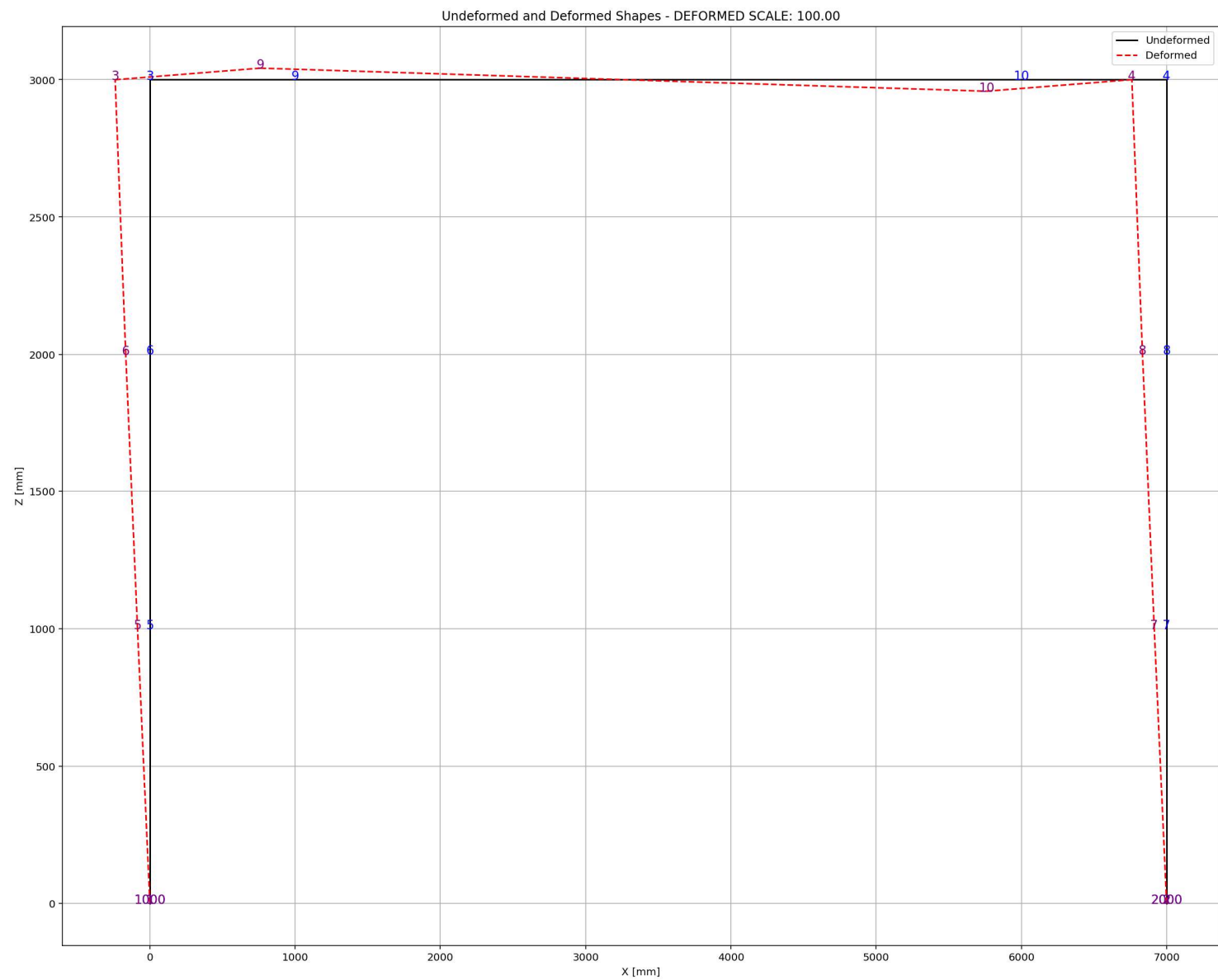
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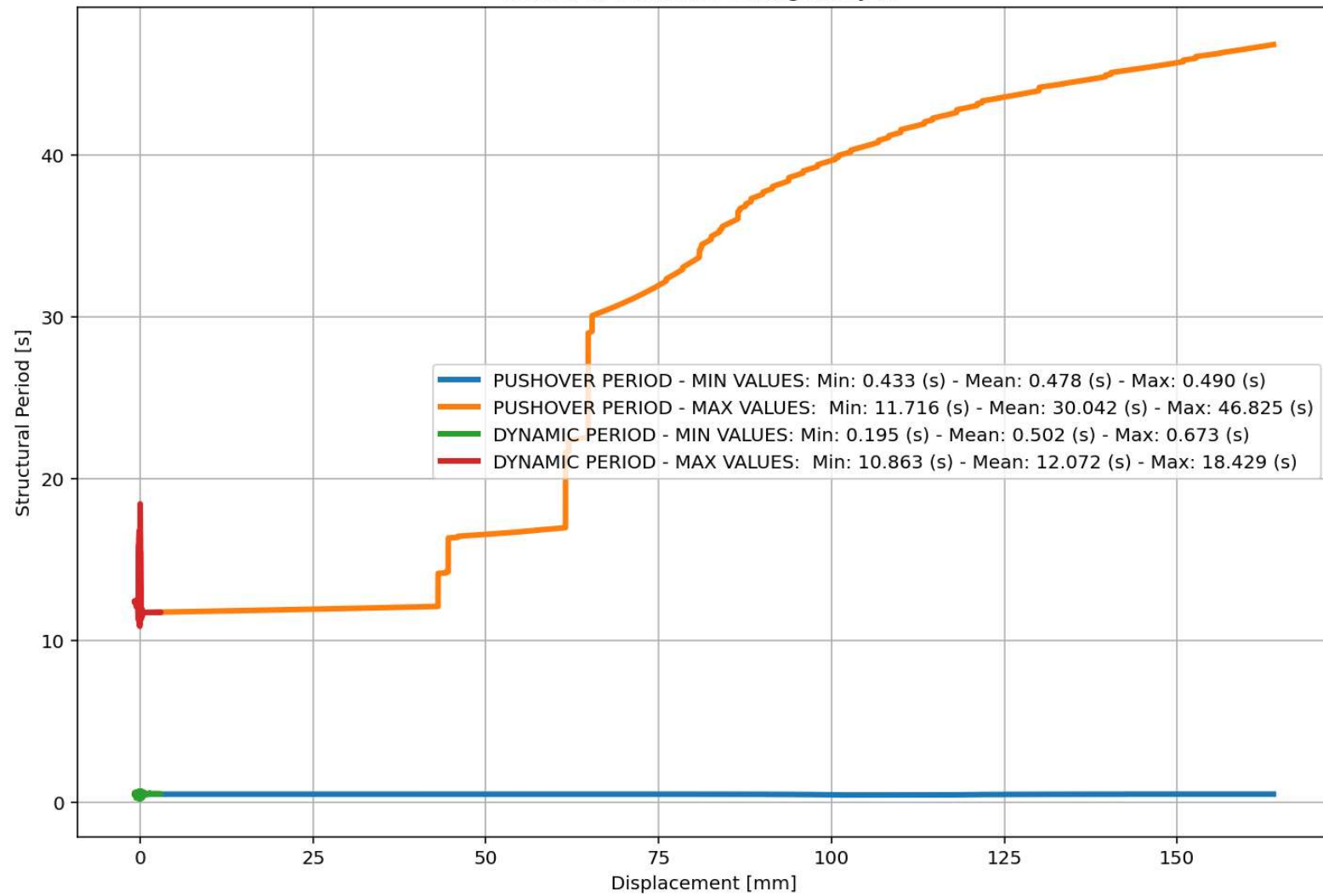
NONLINEAR STATIC ANALYSIS (PUSHOVER)

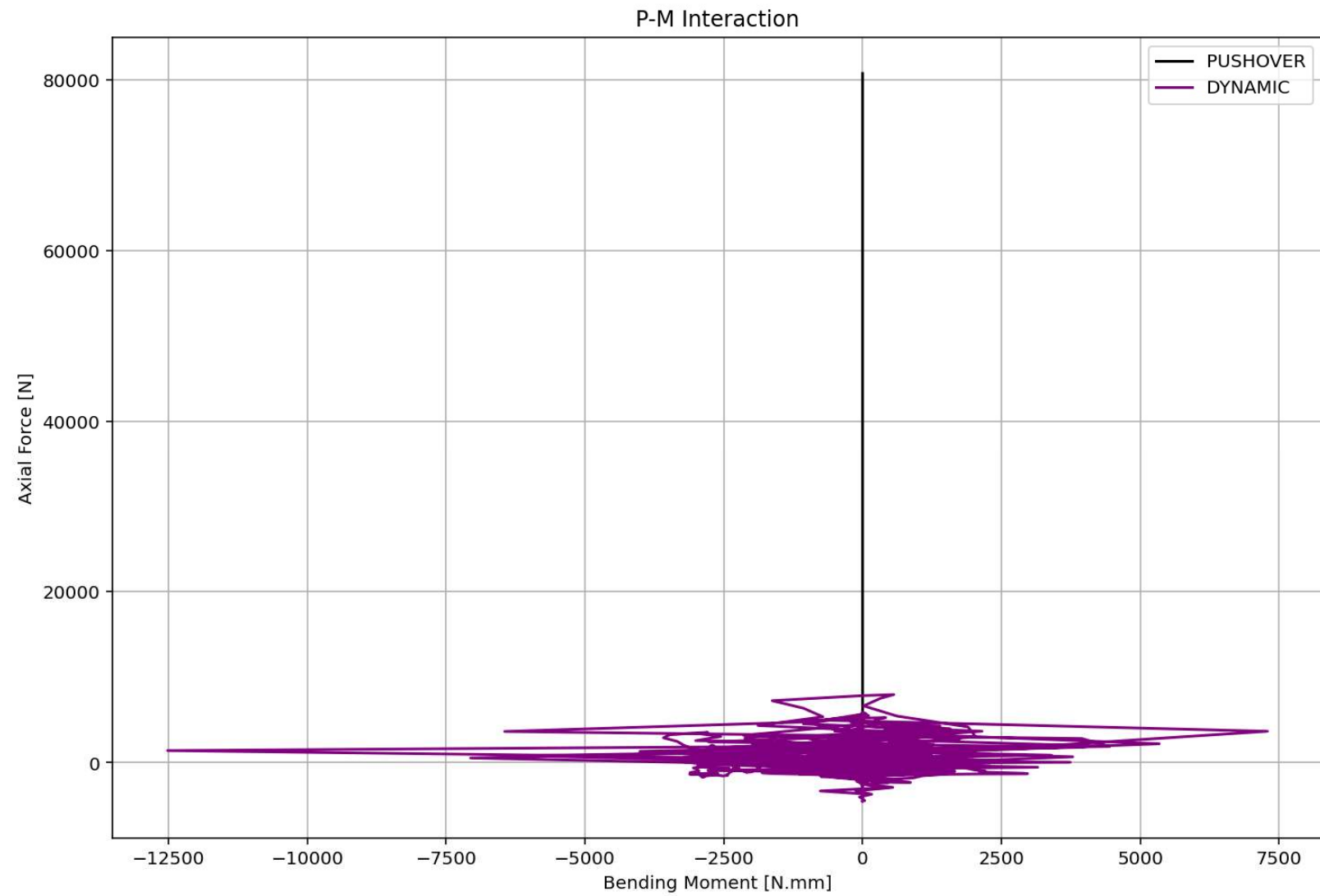


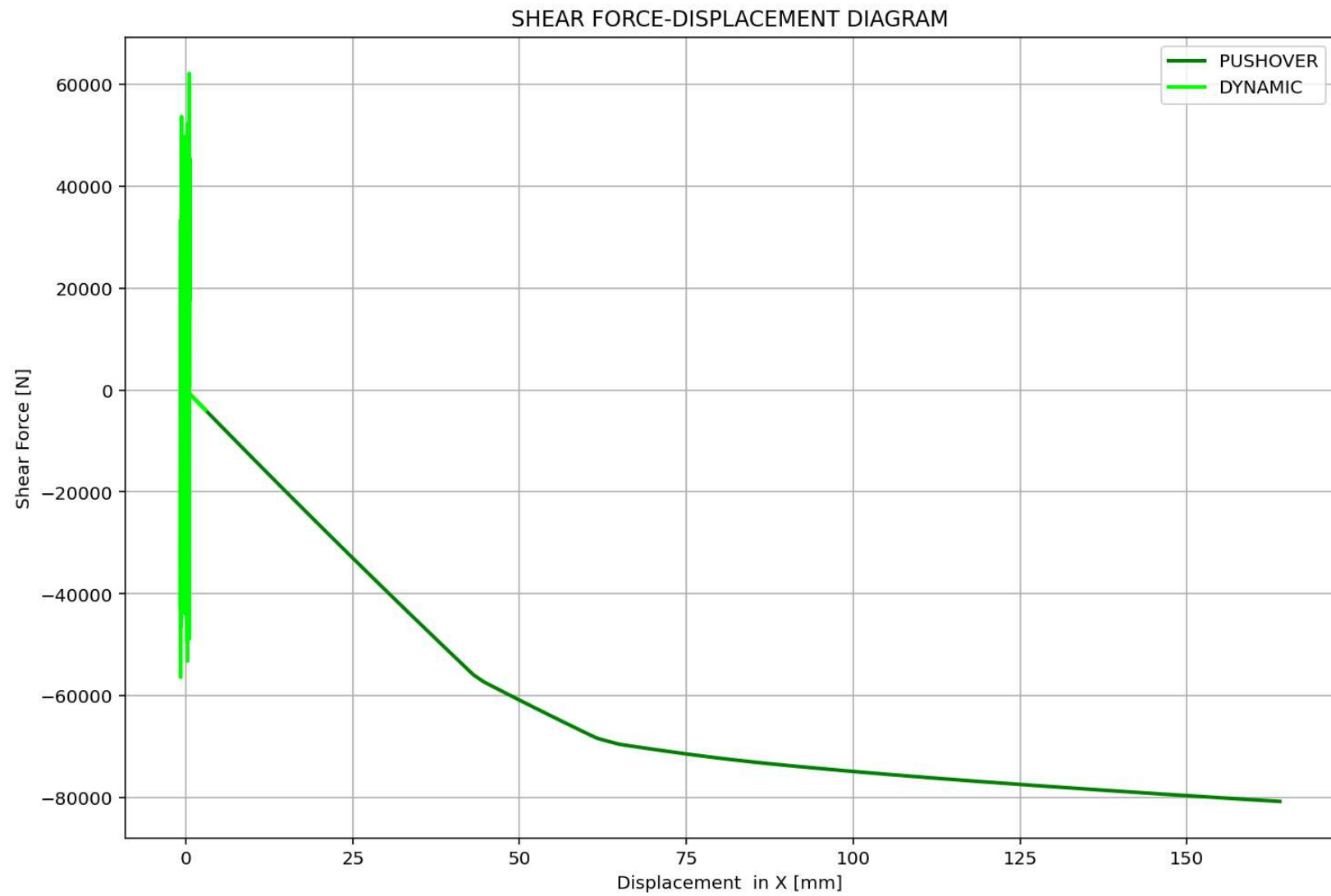
NONLINEAR DYNAMIC ANALYSIS

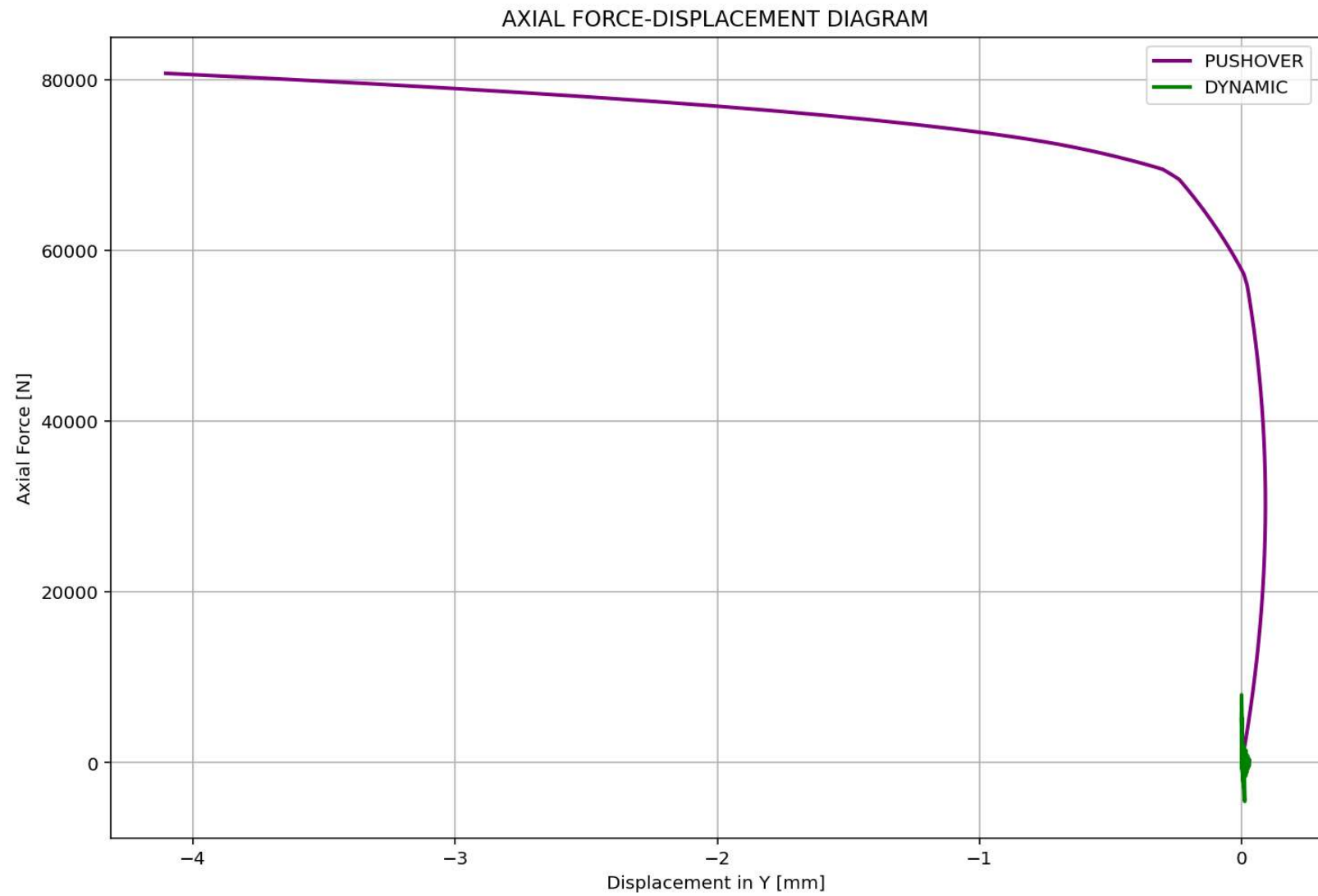


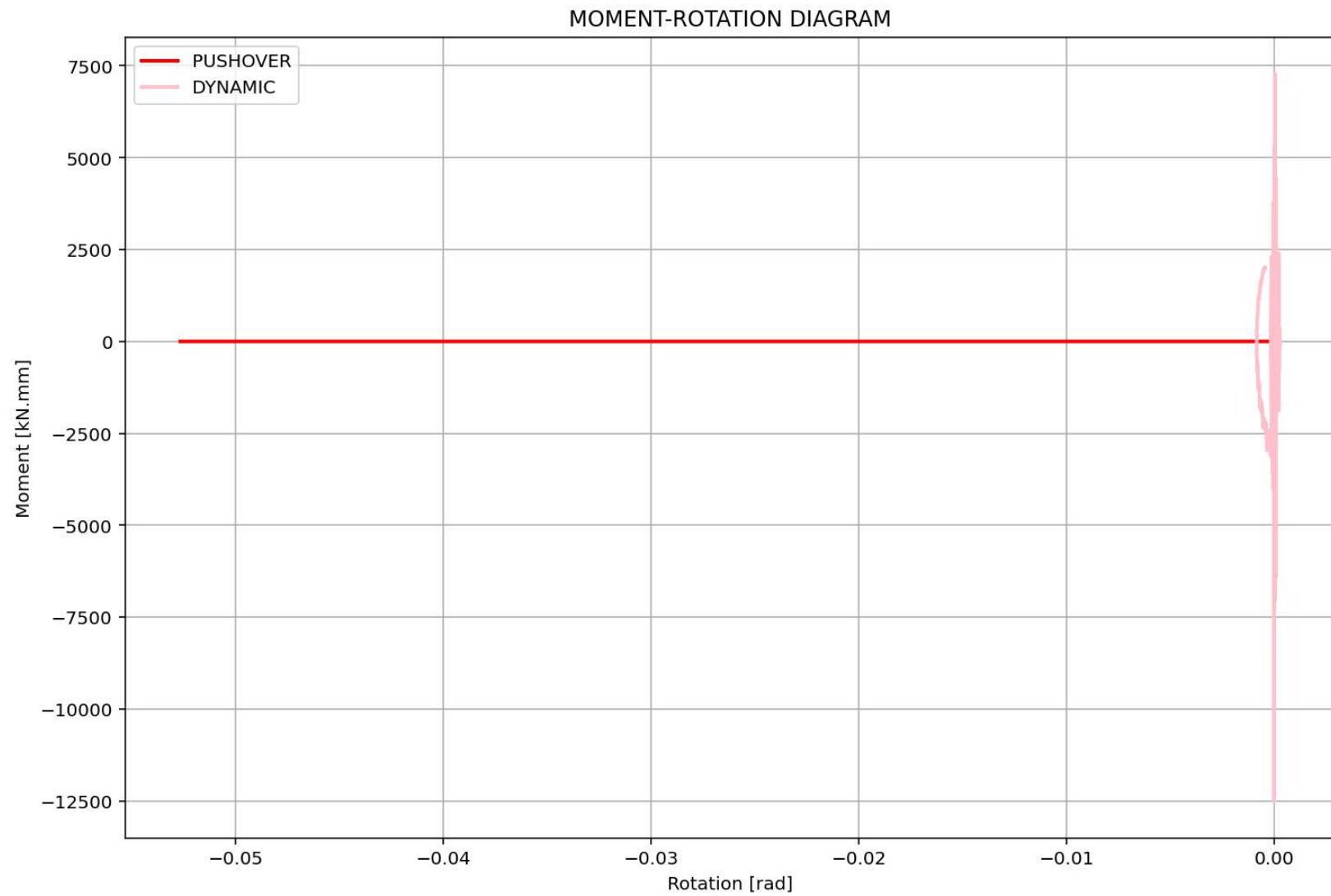
Period of Structure During Analysis

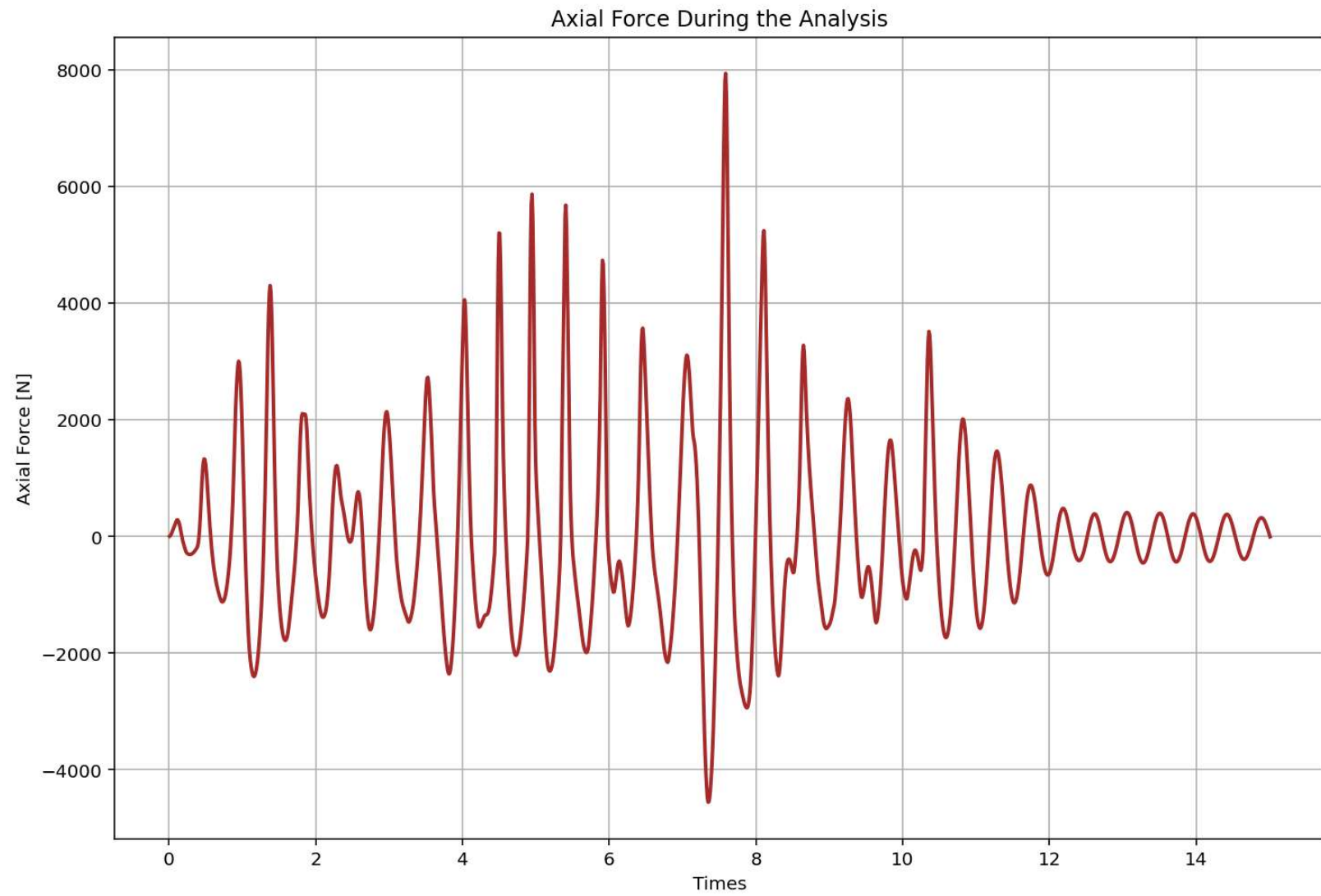


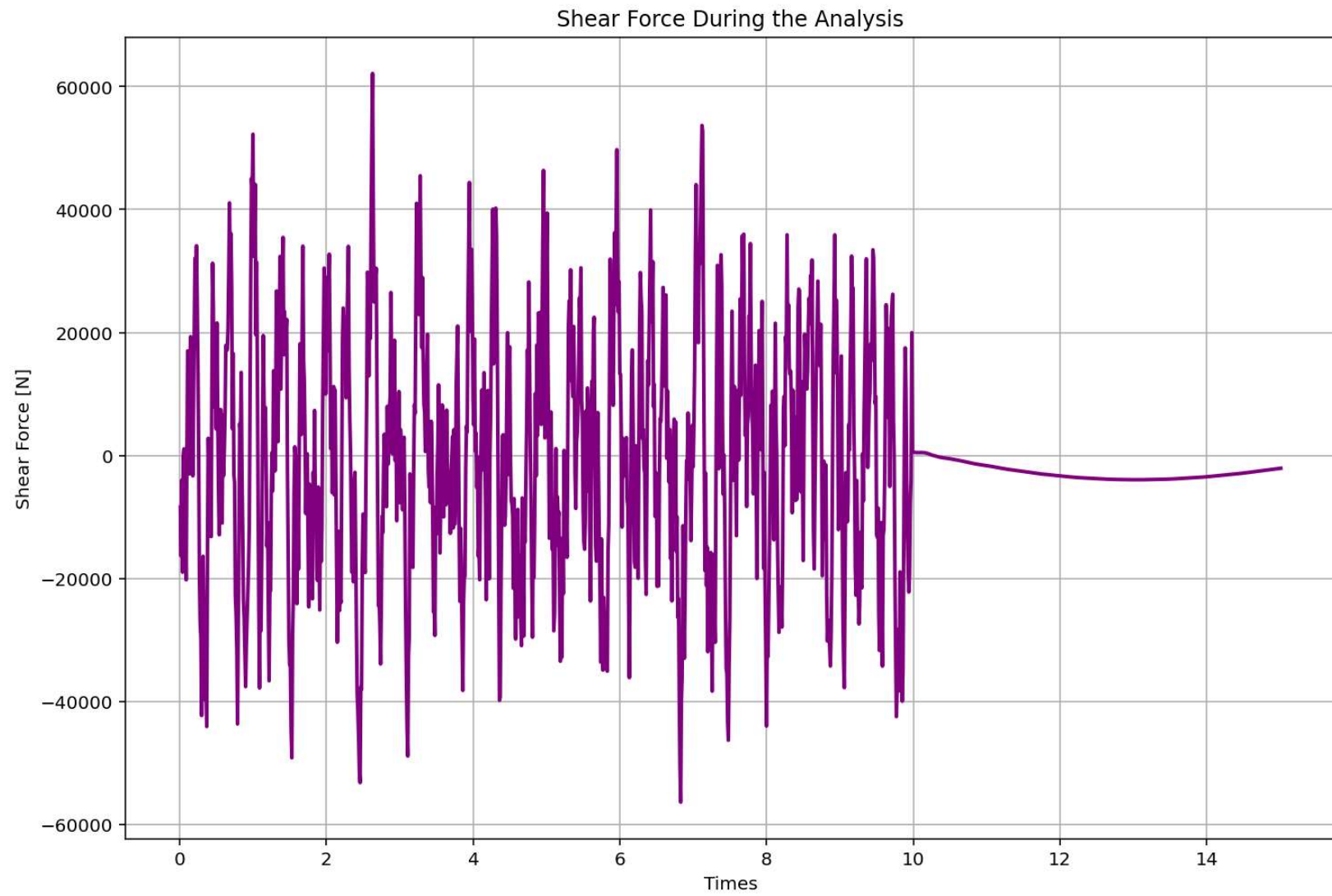


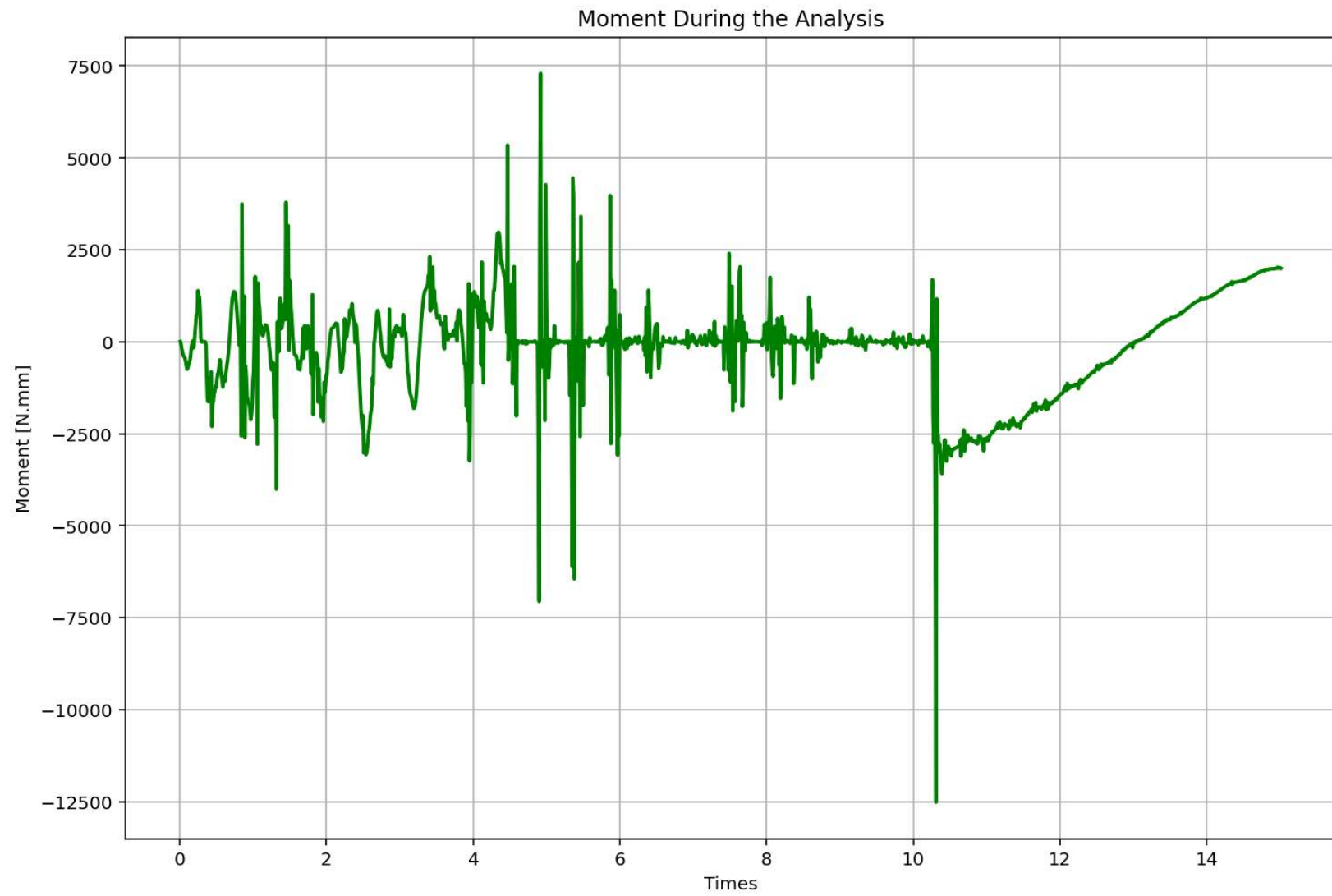


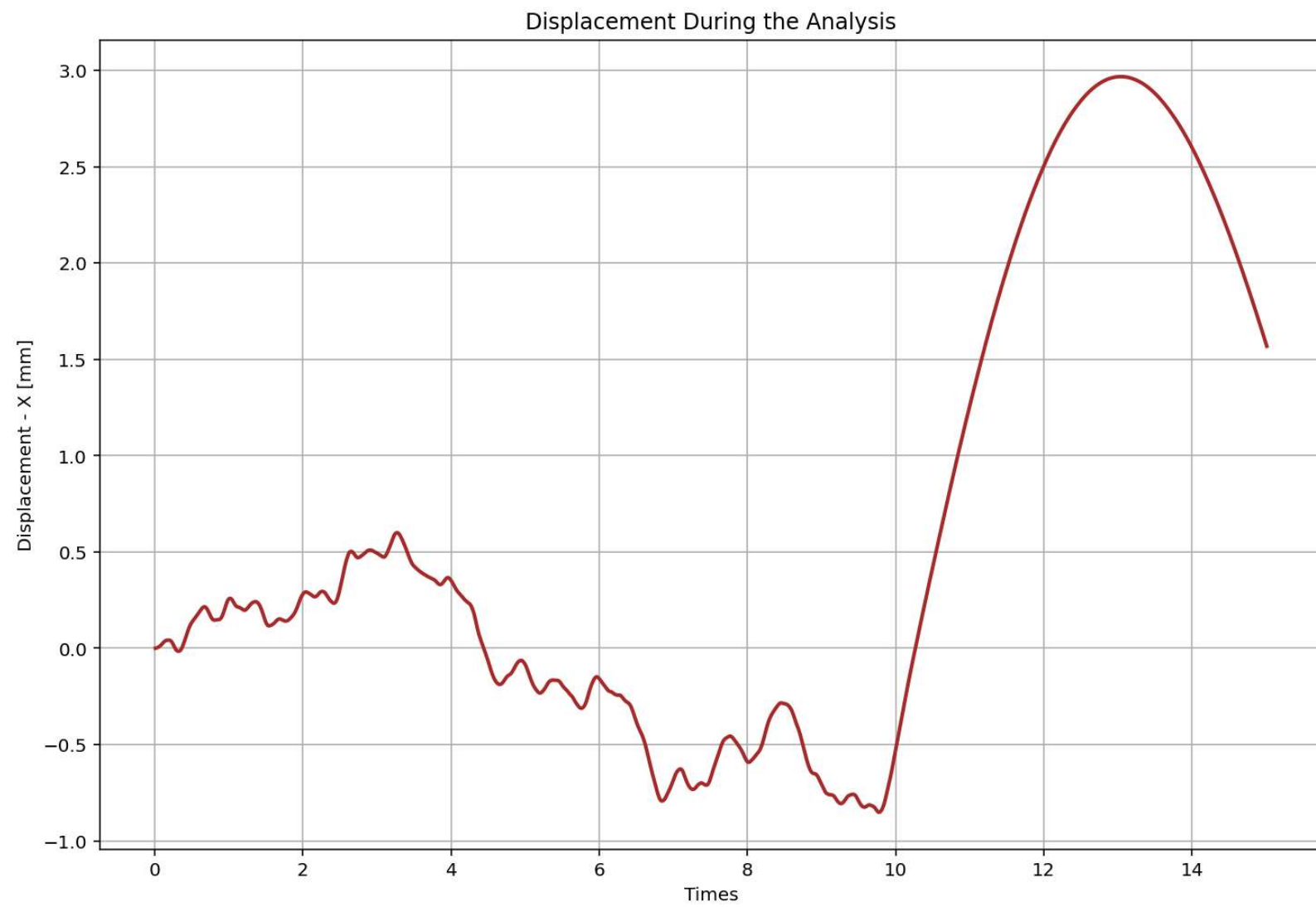


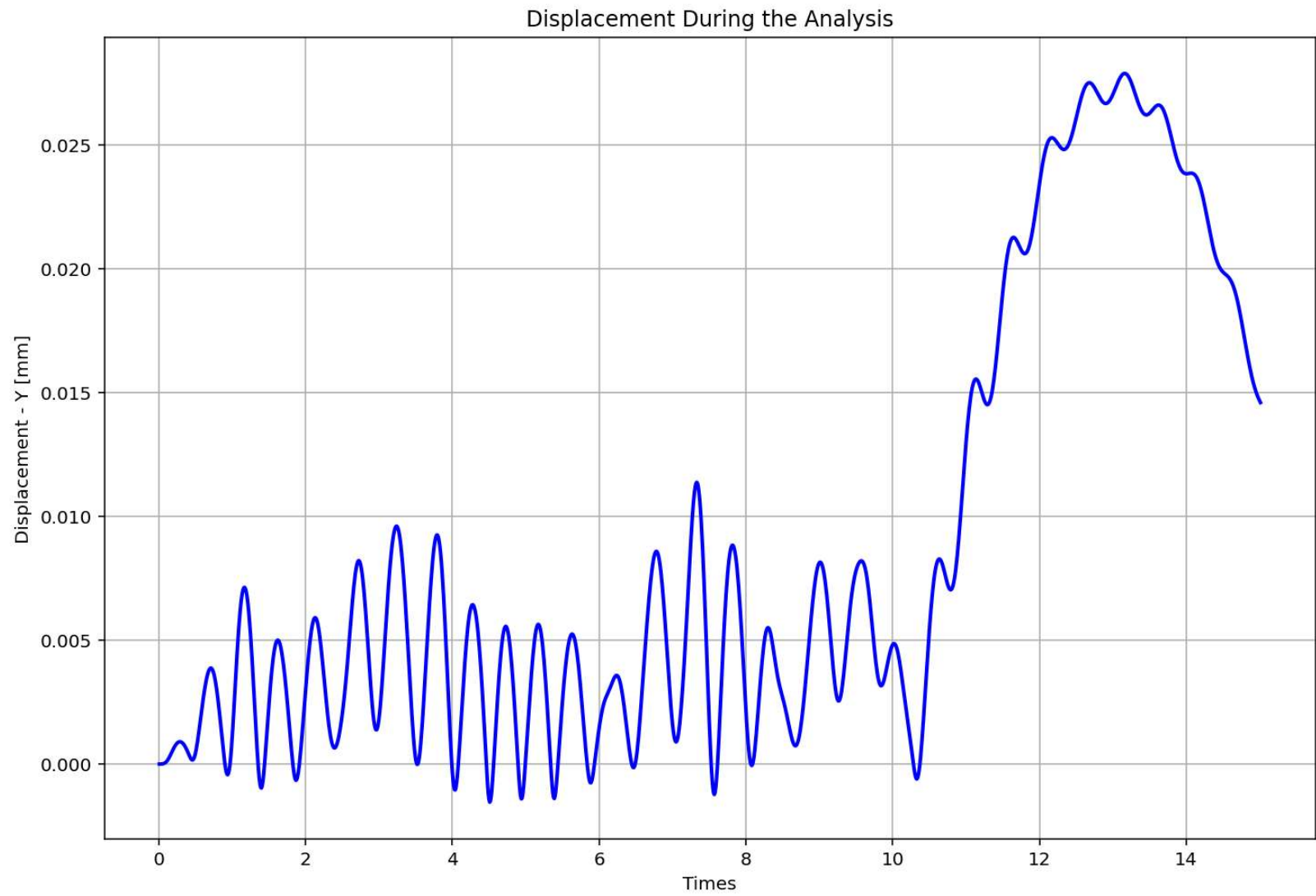


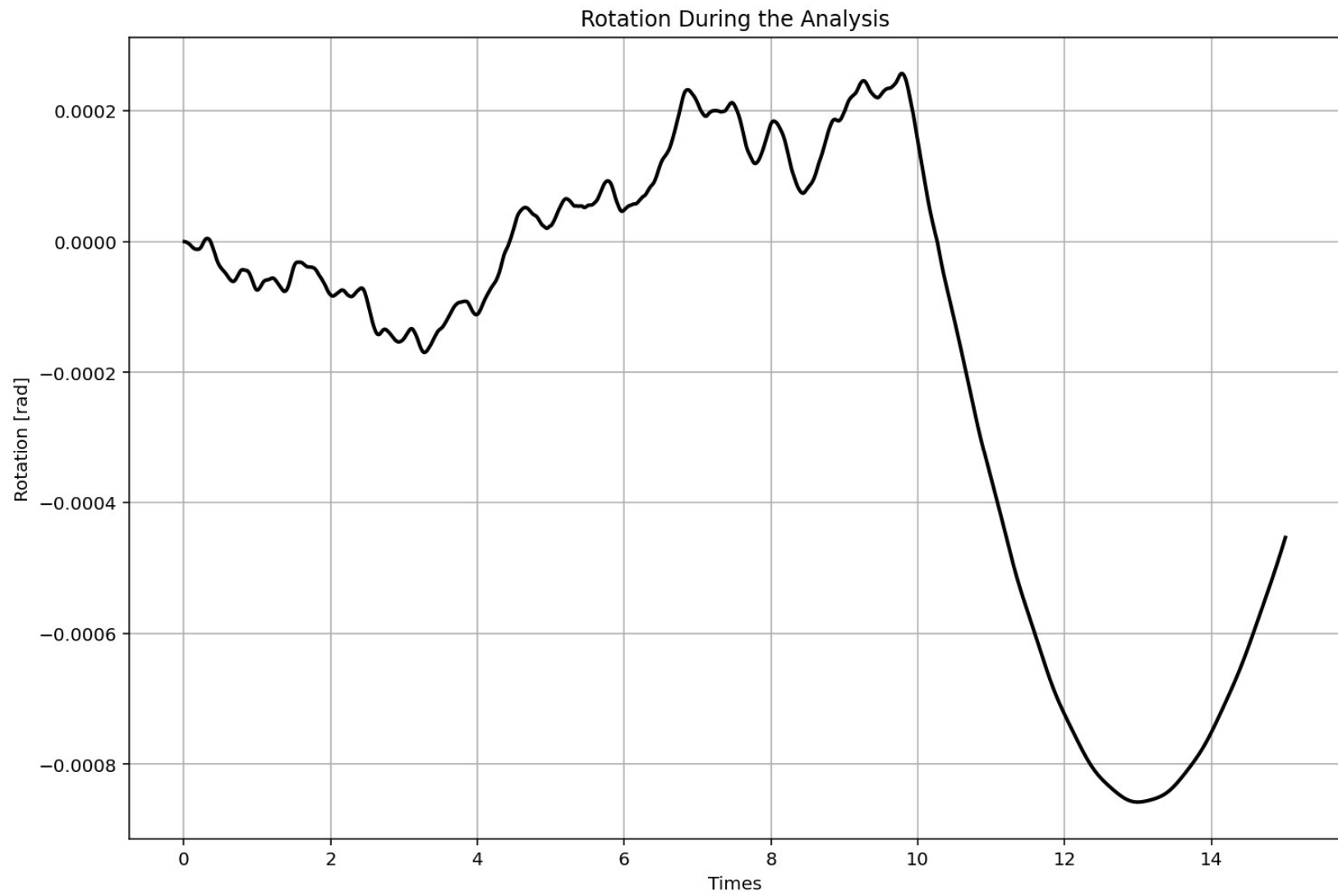




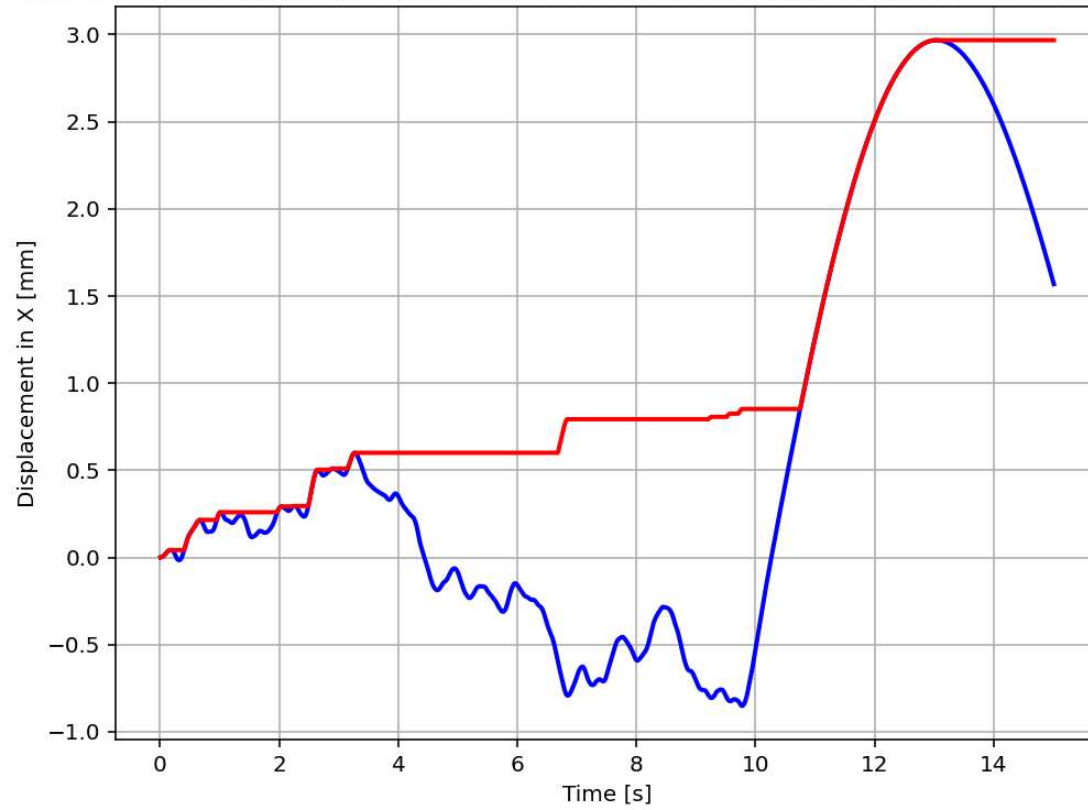




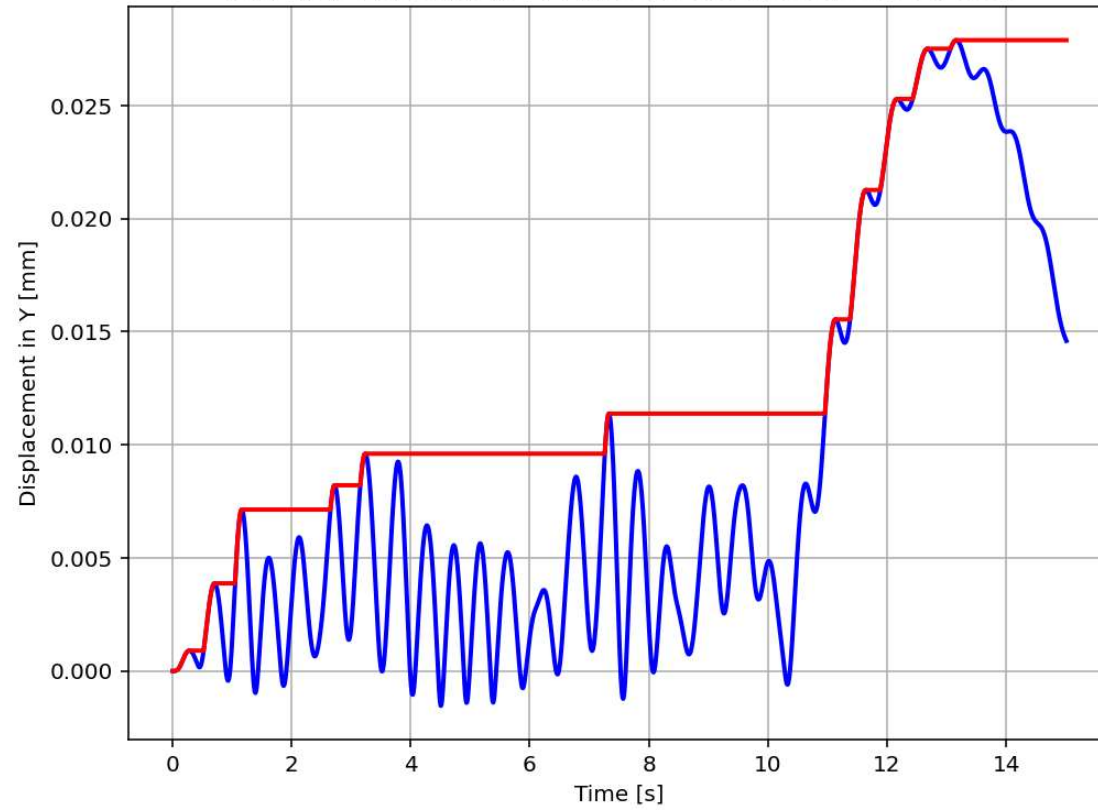




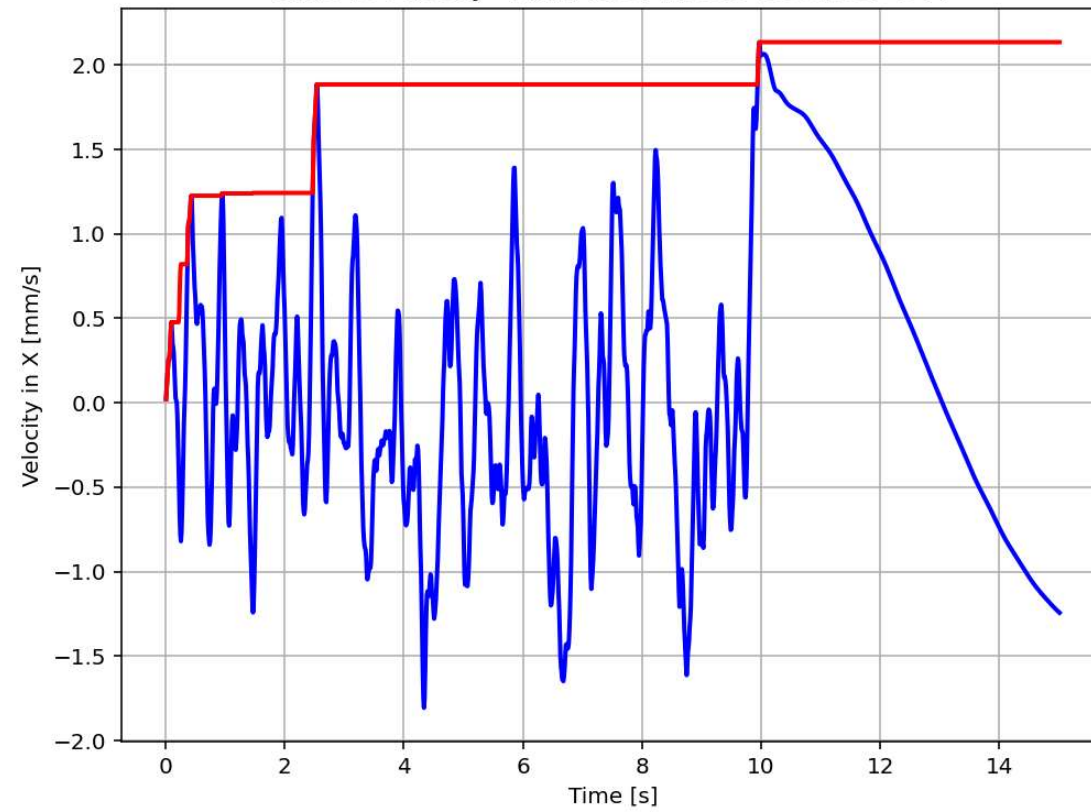
Time vs Displacement - MAX. ABS: 2.967364278127897 | ξ (Calculated): 0.00000e+00 %



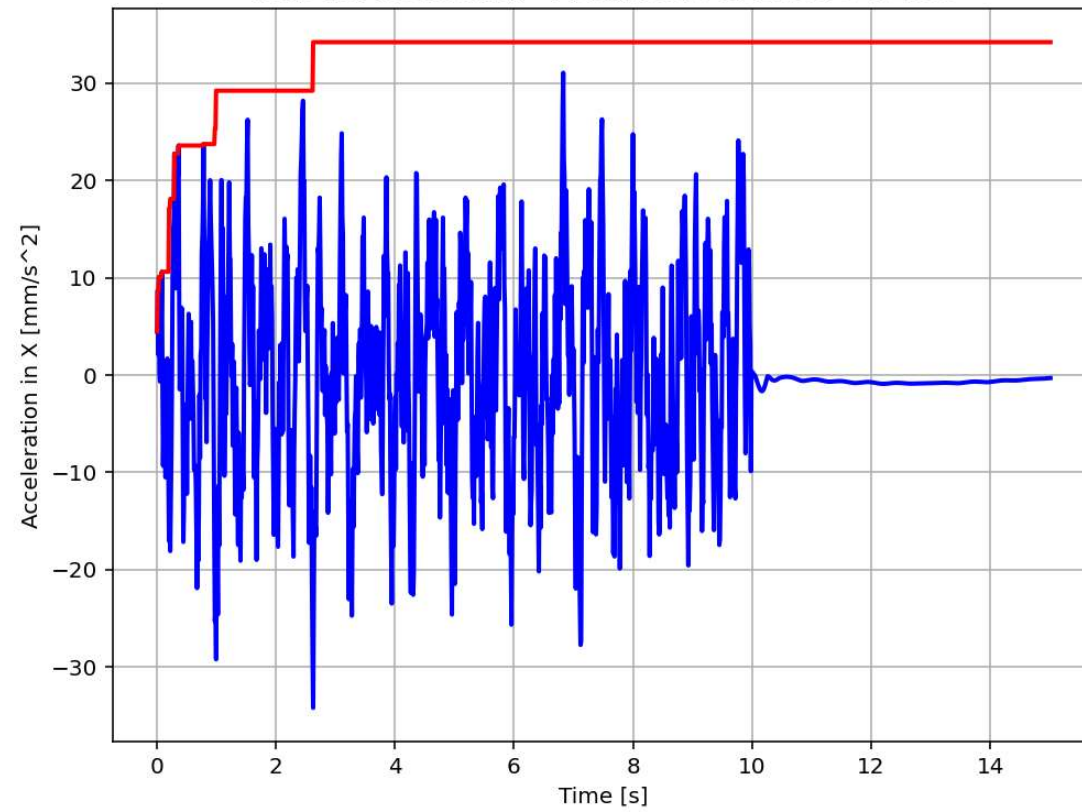
Time vs Displacement - MAX. ABS: 0.027888875851674863

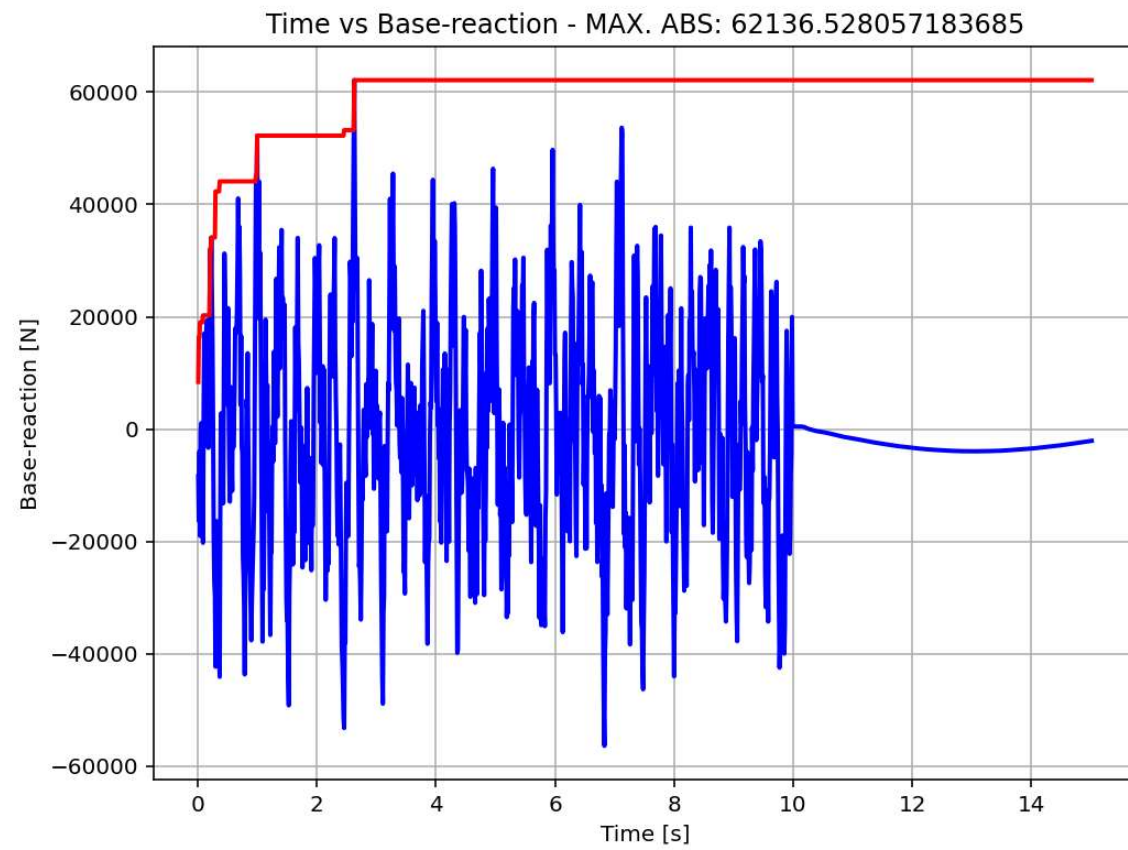


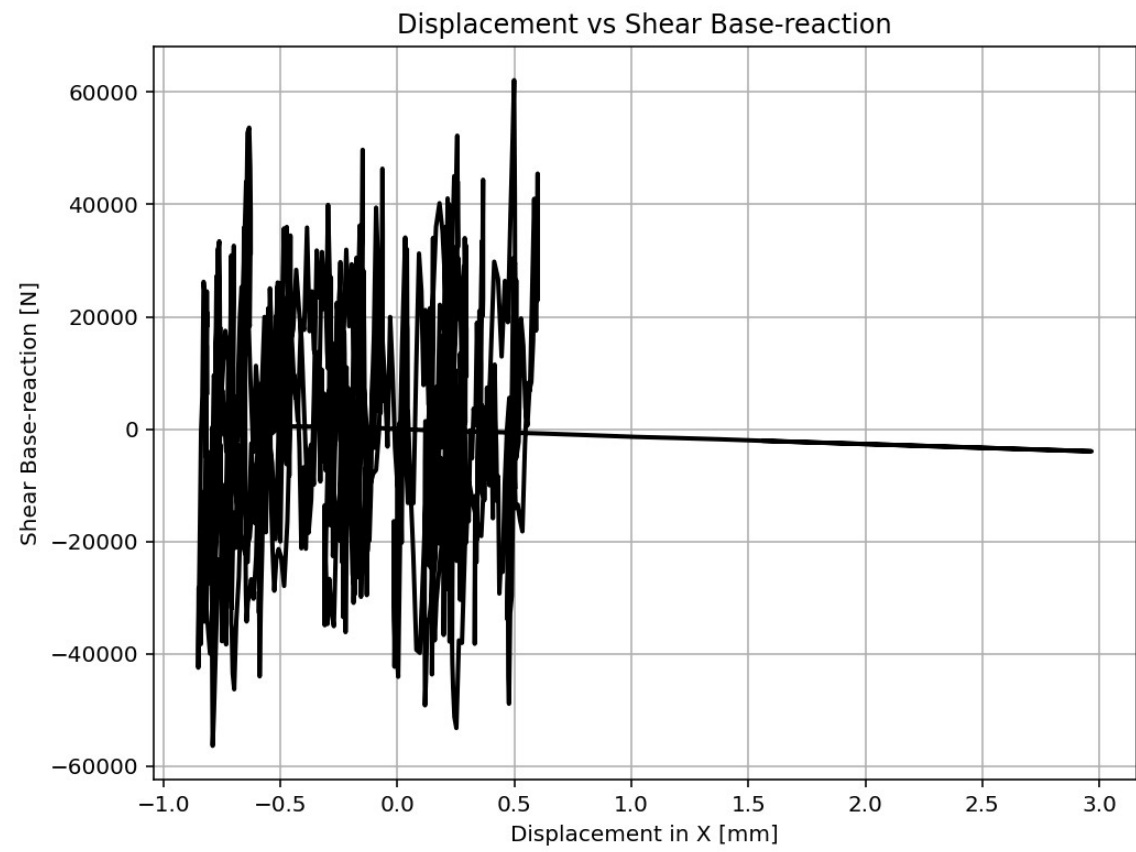
Time vs Velocity - MAX. ABS: 2.1349033022672077

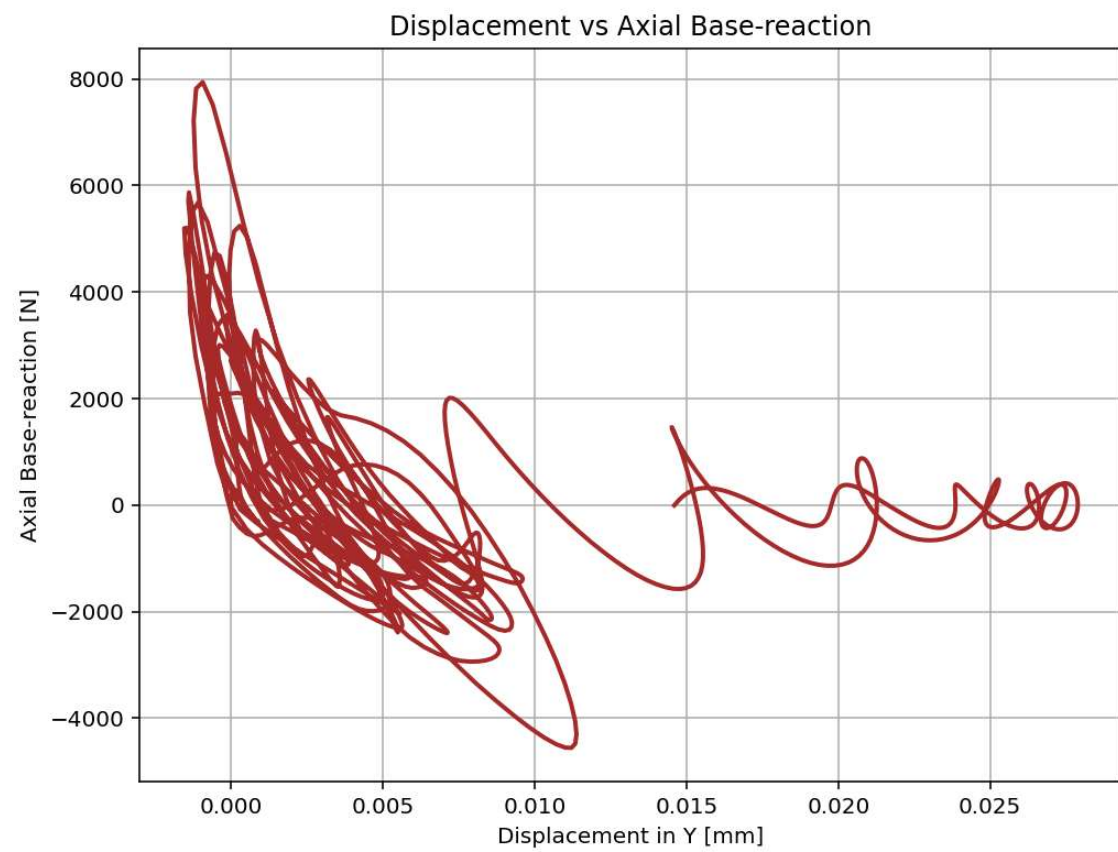


Time vs Acceleration - MAX. ABS: 34.1961179291098

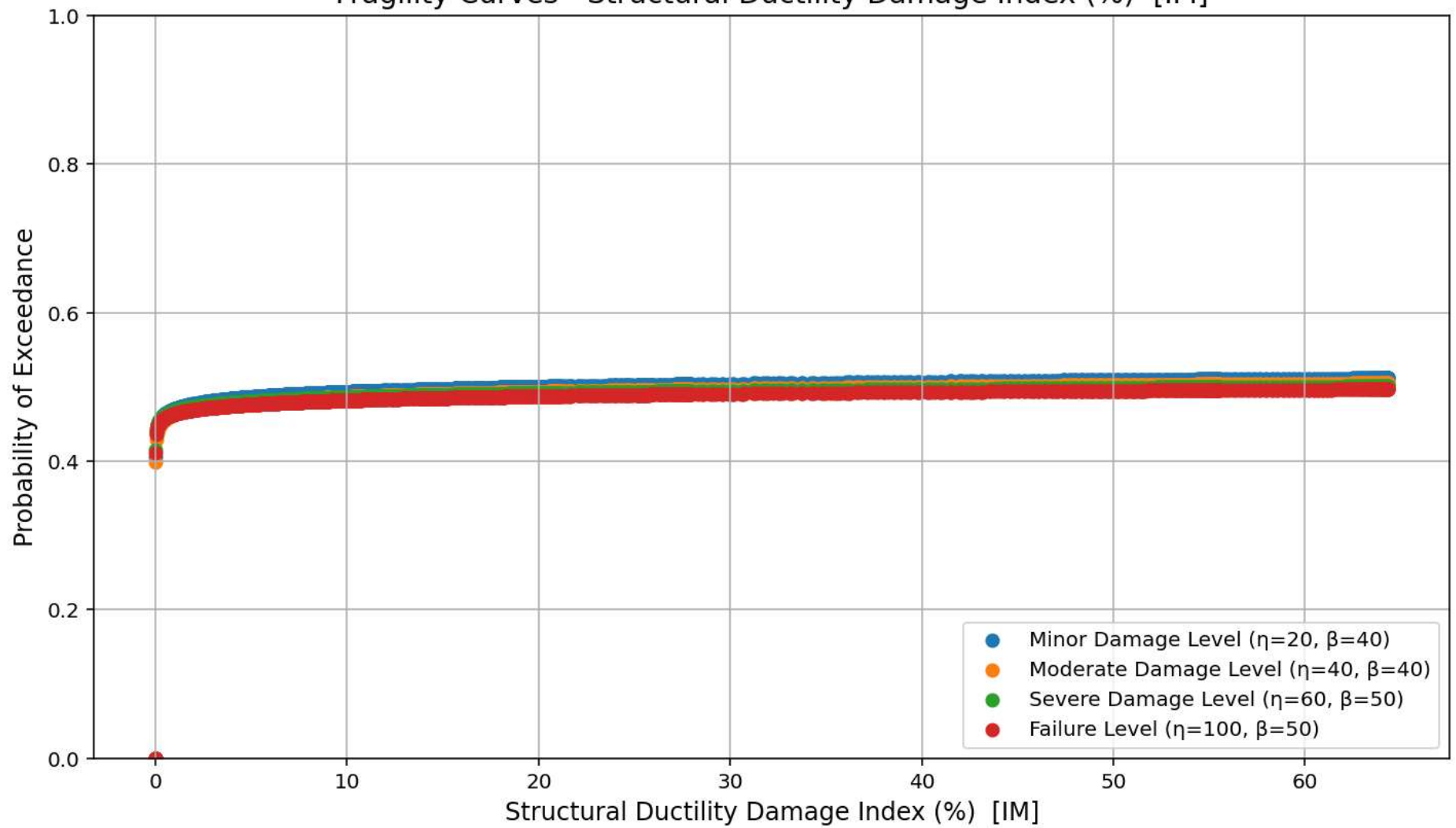




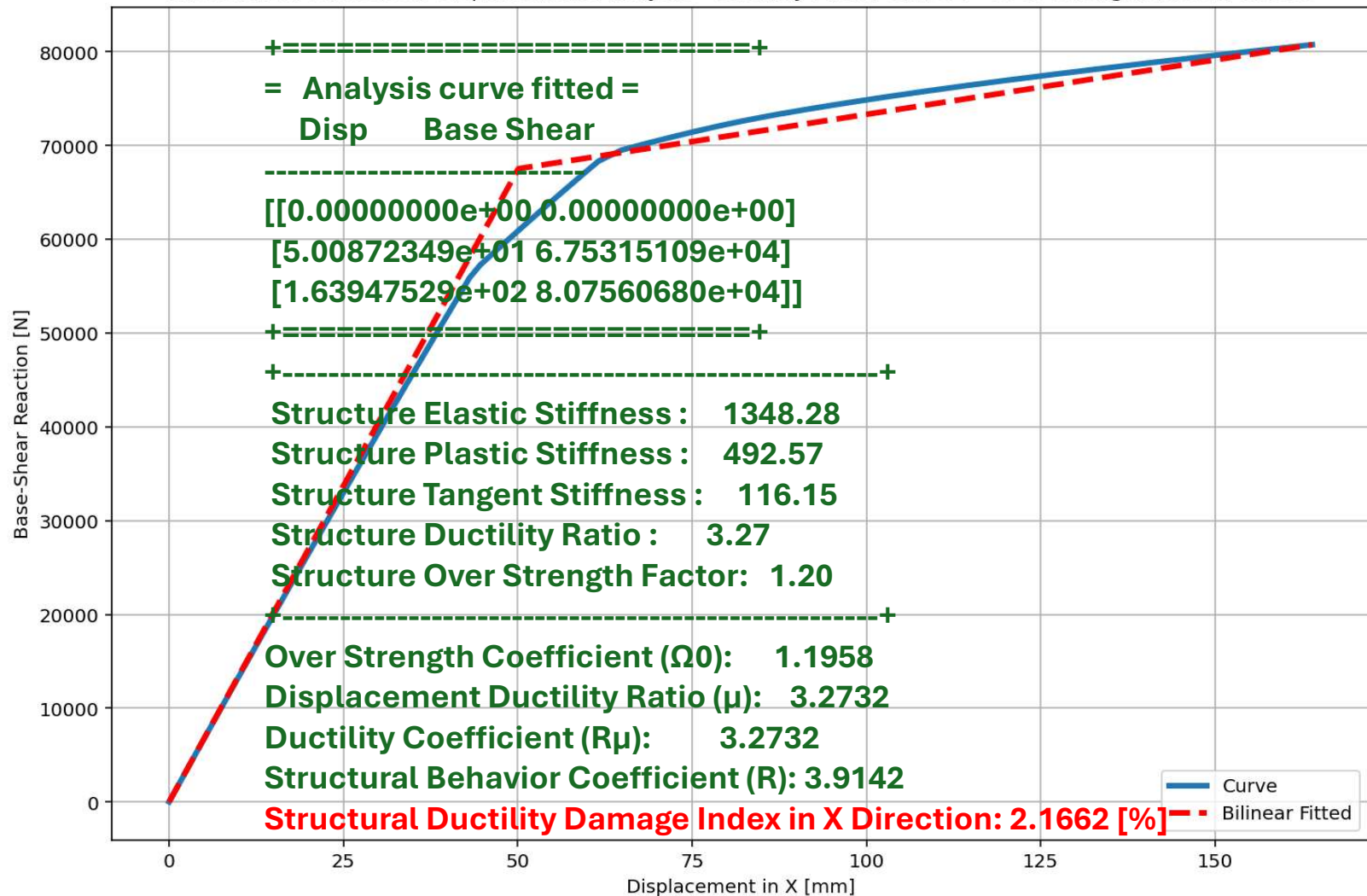




Fragility Curves - Structural Ductility Damage Index (%) [IM]



Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 3.2732 - Over Strength Factor: 1.1958



Last Data of BaseAxial-Displacement Analysis - Ductility Ratio: 9.0305 - Over Strength Factor: 1.0322

